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END SEMESTER EXAMINATION DECEMBER 2024

Name of the Course: B.Tech

Semester: VII

Name of Paper: Software
Verification Validation and
Testing

Paper Code: TCS799

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1/CO5
(a)	Your team identifies a critical defect during the final stages of testing. Which STLC phase might need revisiting, and why?	
(b)	Your company is releasing a safety-critical application. Explain how testing goals and benefits should align with this specific context.	
(c)	A program calculates the square root of an integer x , where $x \in [1, 100]$. Design test cases using BVC, robust testing, and worst-case testing.	
Q2	(20 marks)	CO2/CO4
(a)	Illustrate the bug life cycle with a diagram and explain the transitions between states.	
(b)	A washing machine has three states: Idle, Washing, and Drying. It transitions between states based on events like 'Start', 'Wash Complete', and 'Dry Complete'. Create a state table for this system and design test cases	
(c)	What is decision table-based testing? Explain the significance of rules in a decision table with an example.	
Q3	(20 marks)	CO3/CO6
(a)	Illustrate the Verification and Validation (V&V) model with a diagram. Explain the activities performed at each phase.	
(b)	Compare and contrast Breadth-First and Depth-First Integration Testing in terms of the testing process and outcomes..	
(c)	For the following pseudo-code, calculate the McCabe's Cyclomatic complexity $V(G)$: <pre>if (a > b) { if (c < d) { action1(); } action2(); } else { action3(); }</pre>	
Q4	(20 marks)	CO2/CO4
(a)	You have a program that intermittently crashes with a segmentation fault. Describe the debugging process you would follow to isolate and fix this issue.	
(b)	Discuss the different types of security testing techniques, such as static analysis, dynamic analysis, and black-box testing. Provide examples of when each should be used.	
(c)	What is test case prioritization? Why is it important in software testing?	

Q5	(20 marks)	CO1/CO2
(a)	Compare the use of static and dynamic testing tools in Agile and Waterfall methodologies. Which approach benefits more from each type of tool?	
(b)	What are the primary objectives of automation testing in software development? Name five commonly used automation testing tools and their purposes.	
(c)	List and explain at least three types of UML diagrams commonly used in software testing.	